Mercury

Overview

Mercury is the smallest planet in the Solar System, with a diameter of only 4,879 km — just slightly larger than Earth's Moon. It orbits the Sun at an average distance of 57.9 million km, completing one orbit every 87.97 Earth days, making it the fastest planet in the Solar System with an average orbital speed of 47.87 km/s. Despite being closest to the Sun, Mercury is not the hottest planet; that distinction belongs to Venus, whose thick atmosphere traps heat far more effectively. Mercury has virtually no atmosphere to retain thermal energy, causing dramatic temperature swings: the sun-facing surface reaches 430 °C, while the night side plunges to -180 °C. The planet has a density of 5.43 g/cm^3, second only to Earth, indicating an unusually large iron-rich interior for its size.

Orbit and Rotation

Mercury exhibits one of the most unusual rotation-orbit relationships in the Solar System, known as a 3:2 spin-orbit resonance. For every two orbits Mercury completes around the Sun, it rotates exactly three times on its own axis. As a result, a single solar day on Mercury — the time from one sunrise to the next — lasts 176 Earth days, which is twice its orbital year of 88 Earth days. Mercury also has the most eccentric orbit of any planet, ranging from 46 million km at perihelion (closest approach) to 70 million km at aphelion (farthest point). This extreme orbital eccentricity means that the Sun appears two to three times larger in the sky at perihelion than at aphelion, and the solar energy Mercury receives varies by a factor of more than two across its orbit.

Surface and Geology

Mercury's surface is heavily cratered and resembles the Moon in many respects, preserving a record of the late heavy bombardment period roughly 3.8–4.1 billion years ago. The most prominent surface feature is the Caloris Basin, one of the largest impact craters in the Solar System, spanning approximately 1,550 km in diameter and formed by an asteroid impact around 3.9 billion years ago. The impact that formed Caloris was so powerful that the shock waves travelling through the planet created a region of chaotic, jumbled terrain at the exact antipodal point on the opposite side of Mercury, known as 'weird terrain.' Mercury is also crossed by long, winding cliffs called lobate scarps, some extending for hundreds of kilometres and rising up to 3 km high, formed as the planet's interior cooled and contracted over billions of years. Surprisingly, permanently shadowed craters at Mercury's poles contain water ice, confirmed by NASA's MESSENGER mission, deposited by comets and meteorites over time.

Interior and Magnetic Field

Mercury has a disproportionately large iron core relative to its size, taking up approximately 85 % of the planet's radius and representing about 55 % of its total volume — far larger than the cores of other terrestrial planets. Scientists believe this anomaly may be partly explained by a giant impact early in

Mercury's history that stripped away much of its silicate mantle. Despite its small size and slow rotation, Mercury possesses a global magnetic field — about 1 % as strong as Earth's — generated by convection in its partially molten outer core. This weak magnetosphere is still capable of interacting with the solar wind, creating a tiny magnetosphere and producing brief magnetic tornadoes called flux transfer events that funnel solar wind particles to the surface. The existence of Mercury's magnetic field was a major surprise when Mariner 10 detected it in 1974, because planetary magnetic field models at the time predicted such a small, slow-rotating body could not sustain a dynamo.

Exploration

Mercury is one of the least explored planets in the Solar System, largely because reaching it requires a spacecraft to lose a tremendous amount of orbital energy to 'fall inward' toward the Sun, making missions technically demanding. Mariner 10 was the first spacecraft to visit Mercury, conducting three flybys between 1974 and 1975 and mapping about 45 % of the surface. NASA's MESSENGER spacecraft became the first to orbit Mercury, entering orbit in March 2011 after a seven-year journey involving six planetary flybys to shed enough velocity. Over four years, MESSENGER confirmed the presence of water ice at the poles, revealed the planet's volcanic history, and characterised its unusual magnetic field. ESA and JAXA's joint BepiColombo mission, launched in October 2018, consists of two orbiters and is designed to study Mercury's magnetosphere, surface composition, interior structure, and exosphere in unprecedented detail when it enters orbit.